

7. Importing Data into R

Now that the CSV file is saved in your project folder, we can import it into R.

To **import data** means to bring data from a file into R so we can work with it.

Importing a CSV File

Use this command:

```
data <- read.csv("mydata.csv")
```

Let's break it down.

Part	Meaning
<code>data</code>	The name we are giving the dataset in R
<code><-</code>	Stores something in an object
<code>read.csv()</code>	A function that reads CSV files
<code>"mydata.csv"</code>	The file name

So this line means:

Read the file called `mydata.csv` and store it in an object called `data`.

Tip

The file name must be inside quotation marks.

Looking at the First Few Rows

Use `head()` to see the first few rows:

```
head(data)
```

	student	study_hours	sleep_hours	attendance	quiz_score	major
1	Student 1	5	7	Present	78	Biology
2	Student 2	8	6	Present	91	Business
3	Student 3	3	8	Absent	65	Psychology
4	Student 4	6	7	Present	84	Biology
5	Student 5	7	6	Present	88	Business
6	Student 6	2	5	Absent	60	Psychology

This helps you quickly check that the data imported correctly.

Understanding the Structure

Use `str()` to see the structure of the dataset:

```
str(data)
```

```
'data.frame':  10 obs. of  6 variables:
 $ student      : chr  "Student 1" "Student 2" "Student 3" "Student 4" ...
 $ study_hours  : int   5 8 3 6 7 2 9 4 6 10
 $ sleep_hours  : int   7 6 8 7 6 5 8 6 7 7
 $ attendance   : chr  "Present" "Present" "Absent" "Present" ...
 $ quiz_score   : int  78 91 65 84 88 60 95 72 83 98
 $ major        : chr  "Biology" "Business" "Psychology" "Biology" ...
```

This tells you:

- How many rows there are.
- How many columns there are.
- The names of the columns.
- The type of information in each column.

For example:

- Numbers may be shown as `int` or `num`.
- Words or categories may be shown as `chr`.

Summarizing the Dataset

Use `summary()` to get a quick summary:

```
summary(data)
```

student	study_hours	sleep_hours	attendance
Length:10	Min. : 2.00	Min. :5.0	Length:10
Class :character	1st Qu.: 4.25	1st Qu.:6.0	Class :character
Mode :character	Median : 6.00	Median :7.0	Mode :character
	Mean : 6.00	Mean :6.7	
	3rd Qu.: 7.75	3rd Qu.:7.0	
	Max. :10.00	Max. :8.0	

quiz_score	major
Min. :60.00	Length:10
1st Qu.:73.50	Class :character
Median :83.50	Mode :character
Mean :81.40	
3rd Qu.:90.25	
Max. :98.00	

For number columns, R may show:

- Minimum
- Median
- Mean
- Maximum

For text columns, R may show basic information about the values.

Viewing Column Names

You can see the column names with:

```
names(data)
```

```
[1] "student"      "study_hours" "sleep_hours" "attendance"  "quiz_score"
[6] "major"
```

Plain-Language Explanation

If you run:

```
head(data)
```

you are asking:

Show me the first few students in the dataset.

If you run:

```
str(data)
```

you are asking:

What kind of information is stored in each column?

If you run:

```
summary(data)
```

you are asking:

Give me a quick overview of the dataset.

Practice: Import and Inspect

Practice

Import the dataset and answer these questions:

1. How many columns are in the dataset?
2. What are the column names?
3. What is one number column?
4. What is one category or text column?

Use these commands:

```
data <- read.csv("mydata.csv")
head(data)
str(data)
summary(data)
names(data)
```

Common Mistake

Warning

This will not work if the file is not in your working directory:

```
data <- read.csv("mydata.csv")
```

If you get a file not found error, go back to Section 5 and check your working directory.

Check Your Understanding

Answer these questions:

1. What does it mean to import data?
2. What does `read.csv()` do?
3. Why does `"mydata.csv"` need quotation marks?
4. What does `head(data)` show?
5. What does `summary(data)` show?

Summary

In this section, you learned how to:

- Import a CSV file into R.
- Store the dataset in an object called `data`.
- View the first few rows.
- Check the structure of the dataset.
- Summarize the dataset.
- View column names.